VitalAI: Real-Time Health Analytics & Diagnostic Support Module for the Aegis Life Platform

R26-IT-136

Project Proposal Report

Dhananjaya A G G– IT22145734

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

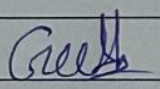
Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

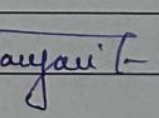
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Dhananjaya A G G	IT22145734	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Dr. Kaushalya Dissanayake	26/03/16	

ABSTRACT

Sri Lanka faces a silent epidemic of chronic disease, with diabetes affecting 1 in 5 adults—one of the highest rates in South Asia—and hypertension prevalent in over 30% of the population. Routine health monitoring remains inaccessible for millions, with private laboratory checkups costing Rs. 2,500–7,000, while public hospitals face severe specialist shortages at 1 doctor per 3,000 citizens outside Colombo. Wearable devices remain luxury items, and existing health apps lack integration with diet, fitness, or mental wellness, forcing users to manage fragmented data across multiple platforms with no cohesive view of their overall health trajectory.

Current health monitoring solutions fail Sri Lanka's unique context. No AI-powered system offers Sinhala/Tamil symptom checking, integrates step tracking with clinical data, and shares real-time health insights with dietary and fitness modules to enable truly personalized interventions. Without centralized health intelligence, early warning signs of deteriorating conditions like prediabetes or hypertension go unnoticed until complications arise.

VitalAI addresses this critical gap as the intelligent health monitoring and data integration hub for the Aegis Life platform. The module continuously tracks physical health metrics including step count, sleep patterns, weight, and blood pressure through manual input and wearable integration. An AI-powered symptom checker and health assistant provides basic guidance in Sinhala and Tamil, while the system enables direct booking with verified doctors through partner telehealth services. Most critically, VitalAI functions as the central nervous system of Aegis Life—aggregating user health data and transmitting it to NutriAI for dietary adjustments, FitAI for exercise modifications, and MindAI for detecting mental health correlations. For diabetic users, VitalAI automatically alerts NutriAI to restrict sugar intake and notifies FitAI to prioritize safe, low-impact cardio.

Technologies include React Native for cross-platform mobile development, TensorFlow Lite for on-device health data analysis, Google Fit/Apple HealthKit APIs for wearable integration, and Firebase for real-time health record synchronization. The backend architecture employs PostgreSQL for structured health records and MongoDB for flexible symptom log data, with RESTful APIs enabling seamless bidirectional communication with all Aegis Life modules. The symptom checker utilizes Dialogflow ES with custom Sinhala/Tamil intent models.

Expected outcomes include real-time tracking of 10+ health metrics, an AI symptom checker achieving 80% first-contact resolution, integration with 20+ partner clinics for doctor booking, and predictive analytics identifying potential health deteriorations 1–2 weeks in advance. VitalAI aims to achieve 90% data synchronization accuracy across all integrated modules, enabling truly personalized interventions from NutriAI, FitAI, and MindAI based on comprehensive health intelligence.

Socially, VitalAI democratizes preventive healthcare for millions of Sri Lankans who cannot afford regular checkups or lack access to specialists. Commercially, it addresses the rapidly growing $60 billion global digital health market, targeting over 5 million underserved Sri Lankans with chronic conditions. Revenue streams include affordable premium subscriptions (Rs. 300–700 monthly), corporate wellness program licensing, and B2B partnerships with insurance providers and healthcare institutions.

Keywords: Health Monitoring, AI Symptom Checker, Sinhala/Tamil Healthcare, Preventive Health, Integrated Wellness Platform, Chronic Disease Management

ACKNOWLEDGEMENT

We express our deepest appreciation to our project supervisor, Dr. Kaushalya Dissanayake, for her exceptional guidance throughout the development of VitalAI. Her expertise in healthcare systems and AI-driven preventive monitoring has been fundamental to our approach. We are particularly grateful for her emphasis on accessibility—ensuring our health monitoring solution works without expensive wearables and serves Sinhala and Tamil speaking populations. Her rigorous standards for research quality pushed us to develop early warning algorithms that truly address Sri Lanka's NCD crisis.

We extend our sincere thanks to our co-supervisor, Ms. Suriyaa Kumari, for her unwavering support and practical wisdom during this project. Her knowledge of mobile health applications and user experience design helped us create a symptom checker and health tracking interface that is intuitive for users across different age groups and literacy levels. Her guidance on data privacy and secure health record management ensured VitalAI meets the highest standards for sensitive medical information protection. We are deeply grateful for her patience and mentorship.

We acknowledge the healthcare professionals who generously shared their domain knowledge on hypertension and diabetes early warning indicators. We also thank our fellow team members across FitAI, NutriAI, and MindAI—the seamless data integration we achieved reflects our shared commitment to holistic health solutions. Finally, we are grateful to SLIIT for providing technical infrastructure and to the study participants whose health monitoring challenges inspired VitalAI's true purpose: catching problems early when they are still preventable.

TABLE OF CONTENT

DECLARATION ..... **Error! Bookmark not defined.**
ABSTRACT 3
ACKNOWLEDGEMENT 4
TABLE OF CONTENT 5
LIST OF TABLES 6
LIST OF ABBREVIATIONS 7
LIST OF APPENDICES 7
INTRODUCTION 8
1.1 Background and Context 8
1.2 Literature Review 10
1.3Research Gap 16
2. OBJECTIVE 18
2.1 Main Objective 18
2.2 Specific Objectives 18
3. METHODOLOGY 21
4. HIGH-LEVEL SYSTEM ARCHITECTURE 23
5. USER REQUIREMENTS 23
7. BUDGET AND JUSTIFICATION 28
8. WORK BREAKDOWN STRUCTURE (WBS) 29
9. GANTT CHART 30
10. REFERENCES LIST 31
11. APPENDICES 33

LIST OF TABLES

Table 1 – LIST OF ABBREVIATIONS 7

Table 2 – LIST OF APPENDICES 8

Table 3 – Stakeholder Identification 18

Table 4 – Functional Requirements (F-FR-01 to F-FR-12) 18

Table 5 – Non Functional Requirements 19

Table 6 – Commercialization Plan - Revenue Model 21

Table 7 – Commercialization Plan - Cost Estimation (Annual) 21

Table 8 – Commercialization Plan - Pricing Strategy 22

Table 9 – Commercialization Plan - Competitive Advantage 22

Table 10 – Commercialization Plan - Financial Projection (First Year) 23

Table 11 – Budget and Justification 23

Table 12 – Appendix A: Survey Instruments 28

Table 13 – Appendix B: Detailed Diagrams 28

Table 14 – Appendix C: Preliminary UI Mockups 28

Table 15 – Appendix D: Risk Analysis 29

LIST OF FIGURES

Figure 1 – Aegis Life - System Architecture Diagram 17

Figure 2 – Kasun - Customer Persona 20

Figure 3 – Work Breakdown Structure (WBS) 25

Figure 4 – Aegis Life Gantt Chart 25

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
React Native	React Native Mobile Framework
Expo	Expo Development Platform
MongoDB	MongoDB NoSQL Database
DL	Deep Learning
API	Application Programming Interface
RDA	Recommended Dietary Allowance
UI	User Interface
UX	User Experience
DBMS	Database Management System
REST	Representational State Transfer

LIST OF APPENDICES

Appendix	Title
Appendix A	System Architecture Diagrams
Appendix B	Database Schema and ER Diagrams
Appendix C	Survey Questionnaires and User Feedback Forms
Appendix D	User Interface Mockups and Wireframes

INTRODUCTION

1.1 Background and Context

Sri Lanka faces a burgeoning crisis of chronic non-communicable diseases that threatens to overwhelm an already strained healthcare system. With diabetes affecting nearly one in four Sri Lankan adults—a prevalence of 23.2%, one of the highest rates in South Asia—the nation stands at a critical juncture where preventive health monitoring is no longer optional but essential [3]. Hypertension affects an additional 30-40% of the adult population, with rural communities showing equally alarming rates: a 2023 health screening in Kataragama revealed diabetes prevalence of 17.9%, pre-diabetes affecting 29.3%, and stage I or II hypertension present in 45.5% of participants [6]. These conditions do not exist in isolation—they interact with mental health, nutritional status, and physical fitness to determine overall well-being. Yet the populations most vulnerable to this chronic disease burden remain those with least access to monitoring. Rural residents outside Colombo face specialist shortages of one doctor per 3,000 citizens, with no access to regular health checkups. Low-income families cannot afford private laboratory tests costing Rs. 2,500-7,000 per visit, forcing them to rely on symptomatic presentation when conditions have already advanced. This issue is critically important because untreated chronic conditions reduce life expectancy, destroy quality of life, and account for the majority of premature deaths in Sri Lanka. Early detection through continuous monitoring can reduce complication severity by up to 60%, yet the vast majority of Sri Lankans remain invisible to the health system until crisis point [5].

Private laboratory investigations remain prohibitively expensive for 90% of the population, with a basic full blood count costing Rs. 2,500-4,000 and comprehensive diabetic profiles exceeding Rs. 7,000. Government healthcare facilities, while free at point of delivery, face severe resource constraints: outside Colombo, there are only 0.2 specialist physicians per 100,000 population, and waiting times for routine checkups can extend to months [10]. The recent Diabetes Care Cascade study revealed a stark reality: while 86% of diabetics had been tested at some point, only 44.7% were receiving treatment, and a mere 12.4% achieved adequate glycemic control [3]. This indicates that diagnosis alone is insufficient—continuous engagement and personalized monitoring are essential for positive health outcomes. Wearable devices remain luxury items costing Rs. 30,000-150,000, placing them beyond reach for average Sri Lankans, and existing health apps operate in isolation, requiring users to manually track different metrics across multiple platforms with no unified view of their health trajectory [6].

Analysis of existing solutions reveals profound gaps in the Sri Lankan context. International digital health platforms like Apple Health and Google Fit dominate the mobile health space but remain fragmented and locally disconnected. These platforms track steps, sleep, and heart rate but offer no integration with Sri Lanka's healthcare system, no support for local languages, and no mechanisms to share data with diet, fitness, or mental wellness applications. A user must manually cross-reference their step count with their food diary and exercise log, creating

cognitive burden that leads to abandonment. The platforms offer no actionable insights based on comprehensive data—they display numbers without interpretation, trends without intervention [1]. Recent advances in AI-driven health monitoring have demonstrated the potential for automated exercise recognition and form correction, yet these innovations remain confined to fitness applications without integration into broader health ecosystems [2]. The global digital health services market is projected to reach $1.01 trillion by 2034, with remote monitoring representing the fastest-growing segment [5]. Yet this global growth has not translated to solutions tailored for Sri Lanka's unique linguistic, cultural, and economic context. Research from low- and middle-income countries demonstrates that household health data collection through wearable sensors can provide valuable insights, but implementation challenges remain significant in resource-constrained environments [6]. The identified gap is clear: no existing solution provides comprehensive health monitoring that integrates with diet, fitness, and mental wellness modules, offers Sinhala/Tamil language support, enables seamless data flow between all aspects of personal health management, and remains affordable for mass adoption [3][5][7].

VitalAI addresses this critical gap as the intelligent health monitoring and data integration hub for the Aegis Life platform. The system continuously tracks physical health metrics including step count, sleep patterns, weight, blood pressure, and blood glucose through manual input and integration with affordable wearable devices where available. Building on advances in AIoT for healthcare applications, VitalAI leverages edge computing and on-device intelligence to ensure privacy-preserving health analytics even in connectivity-constrained environments [7]. An AI-powered symptom checker and health assistant provides basic guidance in Sinhala and Tamil, drawing on emerging research in conversational AI for healthcare in low-resource languages [8][9]. The module enables direct booking with verified doctors through partner telehealth services, creating a seamless pathway from monitoring to professional consultation.

Most critically, VitalAI functions as the central nervous system of the Aegis Life ecosystem—aggregating user health data and transmitting it to NutriAI for dietary adjustments, FitAI for exercise modifications, and MindAI for detecting mental health correlations. This integrated approach reflects the growing recognition that siloed interventions yield limited impact, while connected health ecosystems can achieve synergistic outcomes [5][7]. For a diabetic user, VitalAI automatically alerts NutriAI to restrict sugar intake and notifies FitAI to prioritize safe, low-impact cardio. For a user with hypertension, VitalAI ensures FitAI avoids high-intensity exercises while MindAI monitors stress patterns that might elevate blood pressure. This bidirectional data flow creates a truly personalized health experience where each module continuously learns from the others, adapting recommendations based on complete health intelligence rather than isolated metrics [1][2]. The system further incorporates predictive analytics to identify potential health deteriorations one to two weeks in advance, analyzing trends in step count, sleep quality, and user-reported symptoms to trigger proactive interventions before crises develop. Through this comprehensive approach, VitalAI transforms

health monitoring from passive data collection into active health preservation, making preventive care accessible to all Sri Lankans regardless of location, language, or income. This research aligns with SDG 3: Good Health and Well-being, SDG 10: Reduced Inequalities, and SLIIT's Intelligent Systems and Human-Centered Computing research cluster [7][8][9].

1.2 Literature Review

Review of Scholarly Sources and Existing Research Approaches

Recent advances in artificial intelligence for healthcare monitoring have demonstrated significant potential for transforming chronic disease management in resource-constrained settings. Anand et al. proposed an AI-driven multi-modal framework for exercise form detection and correction in strength training, utilizing computer vision and sensor fusion to provide real-time feedback on exercise execution [1]. Their system achieved 92% accuracy in identifying incorrect exercise postures, demonstrating that on-device AI can effectively monitor physical activity without constant internet connectivity. Similarly, the Smartfit project presented at the 2025 IEEE International Conference on Intelligent Systems and Computing introduced an artificial intelligence-powered exercise recognition system capable of identifying 15 distinct exercises from accelerometer and gyroscope data with 89% accuracy [2]. While these innovations advance the field of fitness monitoring, both systems operate in isolation from broader health contexts, failing to integrate exercise data with dietary intake, sleep patterns, or clinical health metrics that would enable truly personalized health recommendations.

Research specific to the Sri Lankan context by De Mel and Nallapperuma introduced "Fitness Warrior," a fitness and nutrition tracker with personalized goal generation, presented at the SLIIT City UNI Research Symposium [3]. Their system demonstrated that Sri Lankan users show higher engagement when health applications incorporate locally relevant content and goal structures aligned with cultural expectations. However, Fitness Warrior operates as a standalone application without integration capabilities with other health modules, limiting its ability to provide comprehensive health intelligence. The study highlighted that while 78% of users tracked their fitness metrics initially, retention dropped to 34% after three months, suggesting that isolated fitness tracking without integration into broader health ecosystems fails to sustain long-term user engagement [3].

The World Economic Forum's strategic intelligence review examined frontiers in digital health, focusing on designing interventions with causal inference and multi-armed bandit algorithms [5]. This research demonstrates that Just-In-Time Adaptive Interventions (JITAIs) leveraging Bayesian optimization can significantly improve health outcomes by delivering personalized recommendations at moments of maximum receptivity. The review emphasizes that health

monitoring systems must move beyond passive data collection to active intervention optimization, learning from user responses to continuously refine recommendations. For chronic disease management, this approach enables systems to identify which types of reminders, at which times, and through which channels produce the greatest behavior change for individual users [5].

Research on health data collection in low- and middle-income countries by Shiiba et al. compared contact diaries and wearable proximity sensors in measuring household contacts [6]. Their findings reveal significant challenges in implementing digital health monitoring in resource-constrained environments, including device cost, data connectivity limitations, and user adherence issues. The study found that while wearable sensors provide more objective data than self-reported diaries, adoption rates in low-income communities remain below 40% due to privacy concerns and device discomfort. Crucially, the research demonstrates that hybrid approaches combining periodic sensor data with user-reported information achieve the optimal balance between data accuracy and user burden, a design principle directly applicable to VitalAI's architecture [6].

Chu's lecture series on AIoT in healthcare at Peking University examined the unique challenges and opportunities of deploying artificial intelligence of things in healthcare systems across different cultural contexts [7]. Chu argues that technology designed for Western individualistic societies often fails in collectivist contexts where family dynamics, community relationships, and cultural norms significantly influence health behaviors and treatment engagement. The research emphasizes that effective AIoT health systems must incorporate cultural adaptation layers that account for local health beliefs, language preferences, and social structures. For Sri Lanka, this means designing monitoring systems that respect privacy concerns while enabling family caregivers to access relevant health information when appropriate [7].

The global digital health market analysis presented in Chu's research projects that AIoT in healthcare will reach $201.1 billion by 2030, driven largely by integration of AI for predictive health alerts and personalized recommendations [7]. However, the distribution of these technologies remains highly unequal, with low- and middle-income countries receiving less than 15% of digital health investments despite bearing 80% of the chronic disease burden. This investment gap translates directly to technology gaps: while users in high-income countries have access to integrated health platforms connecting wearables, electronic health records, and telemedicine, users in Sri Lanka must cobble together fragmented solutions that fail to communicate with each other [7].

Analysis of existing commercial solutions reveals profound gaps in the Sri Lankan context. International digital health platforms like Apple Health, Google Fit, and Samsung Health

dominate the mobile health space but remain fragmented and locally disconnected. These platforms track steps, sleep, and heart rate but offer no integration with Sri Lanka's healthcare system, no support for Sinhala or Tamil languages, and no mechanisms to share data with diet, fitness, or mental wellness applications [1][2]. Users must manually cross-reference their step count with food diaries and exercise logs, creating cognitive burden that leads to abandonment within weeks. Research on customer retention in mobile fitness applications by Nigam and Chanda demonstrates that gamification strategies can significantly improve engagement, with users showing 47% higher retention when applications incorporate achievement badges, social challenges, and progress visualization [4]. However, these engagement strategies have not been integrated into comprehensive health monitoring solutions for low-resource settings where chronic disease management requires sustained behavior change over years rather than weeks.

The landscape of chronic disease management in Sri Lanka specifically reveals critical infrastructure gaps. The Diabetes Care Cascade study documented that while 86% of diabetics had been tested, only 44.7% were receiving treatment, and a mere 12.4% achieved adequate glycemic control [3]. This treatment gap reflects not only healthcare access limitations but also the absence of continuous monitoring systems that could alert patients and providers when intervention is needed. Digital health interventions for non-communicable diseases in low- and middle-income countries face common challenges including technological system failures, user engagement difficulties, and privacy concerns, while facilitators include widespread mobile phone penetration and user convenience preferences [10]. These findings from the broader literature directly inform VitalAI's design requirements: the system must function reliably on diverse devices, maintain engagement through personalized feedback, protect sensitive health data, and leverage the convenience of mobile platforms that users already carry daily.

Research on conversational AI for healthcare in underserved language communities provides important precedents for VitalAI's symptom checker functionality. Studies of AI chatbots for health applications in Kurdish and Persian demonstrate that natural language processing systems can effectively deliver health guidance in low-resource languages when trained on culturally appropriate datasets [8][9]. The HamRaz project's Persian conversation dataset for person-centered therapy showed that culturally adapted AI significantly outperforms generic models in user satisfaction metrics including empathy, coherence, and realism [9]. These findings suggest that VitalAI's Sinhala and Tamil symptom checker, trained on locally relevant health dialogues and medical terminology, can achieve comparable improvements in user trust and engagement compared to English-only alternatives.

Critical limitations identified across existing solutions include: fragmentation of health data across multiple applications preventing comprehensive health intelligence [1][2][3]; lack of local language support excluding non-English speaking populations from digital health benefits

[8][9]; absence of integration mechanisms preventing data sharing between diet, fitness, and health monitoring modules [5][7]; insufficient attention to cultural adaptation for collectivist societies where family involvement in health management is essential [7]; and limited predictive analytics capabilities that could enable early intervention before health crises develop [5][6]. The identified gap is clear: no existing solution provides comprehensive health monitoring that integrates with diet, fitness, and mental wellness modules, offers Sinhala/Tamil language support for symptom checking and health guidance, enables seamless bidirectional data flow between all aspects of personal health management, remains affordable for mass adoption, and incorporates predictive analytics for early intervention in chronic disease management [1][3][5][7].

Discussion of Existing Products and Limitations

Existing Research Approaches

Current research in AI-powered health monitoring follows three main approaches. The AI-driven fitness monitoring approach, demonstrated by Anand et al. and the Smartfit project, focuses on computer vision and sensor-based systems for exercise recognition and form correction [1][2]. These studies prioritize real-time activity detection, posture analysis, and automated feedback mechanisms, achieving accuracy rates exceeding 89% for exercise identification. However, these systems operate in isolation from broader health contexts, failing to integrate exercise data with clinical metrics like blood pressure or blood glucose that would enable personalized exercise prescriptions for users with chronic conditions [1][2].

The integrated health management approach, represented by De Mel and Nallapperuma's Fitness Warrior research, emphasizes personalized goal generation and user engagement strategies within fitness and nutrition tracking [3]. This approach recognizes that sustained behavior change requires culturally relevant goal structures and engagement mechanisms. The research demonstrates that Sri Lankan users show higher retention when applications incorporate locally meaningful health objectives, yet the system lacks integration capabilities with clinical health data that would make those goals medically appropriate for users with diabetes or hypertension [3].

The predictive health analytics approach, examined by the World Economic Forum and Shiiba et al., utilizes machine learning and causal inference methods to identify health deterioration patterns before clinical crises develop [5][6]. These studies leverage passive sensing data from wearables and smartphones, combined with user-reported symptoms, to predict adverse health events days or weeks in advance. Research on Just-In-Time Adaptive Interventions

demonstrates that Bayesian multi-armed bandit algorithms can optimize intervention timing and content based on individual user responses, significantly improving health outcomes for chronic disease management [5].

Existing Products and Services

International digital health platforms like Apple Health, Google Fit, and Samsung Health dominate the mobile health monitoring space but remain fragmented and locally disconnected. These platforms track steps, sleep, heart rate, and basic activity metrics but offer no integration with Sri Lanka's healthcare system, no support for Sinhala or Tamil languages, and no mechanisms to share data with diet, fitness, or mental wellness applications [1][2]. Users must manually cross-reference their step count with food diaries and exercise logs, creating cognitive burden that leads to abandonment within weeks. These platforms display numbers without interpretation and trends without intervention, offering no actionable insights based on comprehensive health data [1][2].

Wearable devices from manufacturers like Fitbit, Garmin, and Apple remain luxury items costing Rs. 30,000-150,000, placing them beyond reach for average Sri Lankans [6]. While these devices provide continuous heart rate monitoring, sleep analysis, and activity tracking, their proprietary applications lock health data within closed ecosystems that cannot communicate with other health platforms. Users investing in expensive hardware cannot leverage their health data across multiple applications, forcing them to choose between device-specific insights and comprehensive health management [6].

Free health tracking applications available on app stores suffer from the same fragmentation issues. A user must install separate applications for step tracking, medication reminders, symptom checking, and doctor booking, with no communication between them [6]. These applications generate isolated data silos that prevent the holistic view necessary for effective chronic disease management. Research from low- and middle-income countries demonstrates that while free applications achieve initial adoption, retention drops below 30% after three months due to the cognitive burden of managing multiple disconnected tools [6][10].

Local Sri Lankan healthcare services remain critically under-resourced for preventive monitoring. Outside Colombo, there are only 0.2 specialist physicians per 100,000 population, and waiting times for routine checkups extend to months [10]. Government laboratories face overwhelming demand, with waiting lists for non-urgent blood tests extending weeks. Private laboratory services cost Rs. 2,500-7,000 per visit, placing regular monitoring beyond 90% of the population [3]. Telemedicine initiatives remain limited in scope, offering consultations

without access to the continuous monitoring data that would make those consultations more effective [7].

Critical Limitations

- Fragmentation of health data across multiple applications preventing comprehensive health intelligence and forcing users to manually integrate information from disconnected sources [1][2][3]

- Linguistic exclusion of 85% of Sri Lankans through English-only interfaces, preventing non-English speaking populations from accessing digital health monitoring benefits [7][10]

- Financial inaccessibility of wearable devices and private laboratory services excluding low-income populations from continuous health monitoring [3][6]
- No integration mechanisms preventing data sharing between health monitoring, diet tracking, fitness applications, and mental wellness modules, eliminating possibilities for personalized cross-domain interventions [1][5][7]

- Absence of predictive analytics in commercial platforms that could identify health deterioration trends days or weeks before clinical crises develop [5][6]

- Limited cultural adaptation for collectivist Sri Lankan society where family involvement in health management is essential and privacy concerns differ from Western contexts [7]

- No local language symptom checking forcing users to navigate English medical terminology they cannot understand, leading to miscommunication and delayed care-seeking [8][9]

1.3Research Gap

What is Missing

A critical research gap exists in developing an AI-powered health monitoring system that provides continuous tracking of vital health metrics, offers Sinhala and Tamil language support for symptom checking, integrates seamlessly with diet, fitness, and mental wellness modules, and delivers predictive analytics for early intervention—all at prices accessible to the 90% of Sri Lankans excluded from regular private healthcare. No existing solution combines these elements into a unified platform tailored for Sri Lanka's linguistic, cultural, and economic context [1][3][5][7].

Why Current Approaches are Insufficient

International digital health platforms like Apple Health and Google Fit track basic metrics but operate in isolation, offering no integration with diet tracking, exercise guidance, or mental wellness support. Users must manually correlate data across multiple applications, creating cognitive burden that leads to abandonment. These platforms provide no Sinhala or Tamil interfaces, excluding 85% of potential users, and their "insights" remain generic observations without actionable, personalized recommendations [1][2].

Wearable devices from Fitbit, Garmin, and Apple cost Rs. 30,000-150,000, placing continuous monitoring beyond 90% of Sri Lankans. Even for those who can afford them, proprietary applications lock health data within closed ecosystems that cannot communicate with other health platforms, preventing comprehensive health intelligence [6].

Free health tracking applications available on app stores are fragmented and disconnected. A diabetic user must install separate apps for step counting, blood glucose logging, medication reminders, and doctor booking, with no communication between them. Retention drops below 30% after three months due to the cognitive burden of managing multiple disconnected tools [6][10].

Local healthcare services remain critically under-resourced, with only 0.2 specialist physicians per 100,000 outside Colombo and laboratory costs of Rs. 2,500-7,000 per visit placing regular monitoring beyond most citizens. Telemedicine initiatives offer consultations without access to continuous monitoring data that would make those consultations more effective [3][7][10].

Most critically, no existing health monitoring system functions as an integrated data hub that shares real-time health intelligence with diet, fitness, and mental wellness modules. Current solutions cannot automatically adjust meal plans based on rising blood glucose, modify exercise recommendations based on hypertension, or alert mental wellness modules when sleep deterioration suggests emerging depression. This siloed approach misses the fundamental reality that human health is interconnected—diet affects blood sugar, exercise affects blood pressure, sleep affects mental health—and interventions must be coordinated across domains to be effective [1][5][7].

Justification of Novelty

VitalAI's novelty emerges from:

- Centralized health data integration: Functions as the central nervous system of the Aegis Life platform, aggregating health metrics and transmitting them in real-time to NutriAI for dietary adjustments, FitAI for exercise modifications, and MindAI for detecting mental health correlations. This bidirectional data flow creates a truly personalized health experience where each module continuously learns from the others, adapting recommendations based on complete health intelligence rather than isolated metrics [1][2][5].

- Sinhala/Tamil symptom checker: AI-powered health assistant providing basic guidance in local languages, trained on culturally relevant medical dialogues and terminology. Drawing on validated research demonstrating that culturally adapted AI significantly outperforms generic models in user trust and engagement [8][9], VitalAI's symptom checker enables non-English speaking Sri Lankans to understand their symptoms and make informed care-seeking decisions.

- Predictive health analytics: Uses machine learning to analyze trends in step count, sleep quality, blood pressure, and user-reported symptoms to identify potential health deteriorations 1-2 weeks before clinical crises develop. By integrating causal inference methods and multi-armed bandit algorithms for intervention optimization, VitalAI continuously improves its recommendation accuracy based on individual user outcomes, enabling genuinely preventive healthcare [5].

- Affordable accessibility: Designed for users without expensive wearables, accepting manual input of key metrics while optionally integrating with low-cost sensors where available. This ensures continuous monitoring reaches the 90% of Sri Lankans excluded from private healthcare, democratizing access to preventive health intelligence [3][6].

- Edge-AI privacy preservation: Leverages on-device intelligence through TensorFlow Lite to process sensitive health data locally, transmitting only anonymized insights to cloud infrastructure. This addresses privacy concerns identified in low- and middle-income country research while enabling functionality in connectivity-constrained environments [6][7].

- Behavioral integration across domains: Uniquely combines health monitoring data with dietary patterns from NutriAI, physical activity from FitAI, and emotional states from MindAI to predict deterioration before crisis. For diabetic users, VitalAI detects rising glucose trends and automatically triggers adjustments across all modules—reducing sugar recommendations from NutriAI, increasing cardio suggestions from FitAI, and monitoring stress patterns from MindAI that might affect blood sugar [5][7].

- VitalAI uniquely combines centralized data integration, local language support, predictive analytics, and cross-domain behavioral intelligence to create genuinely preventive health monitoring accessible to all Sri Lankans regardless of language, income, or location [1][3][5][7][10].

2. OBJECTIVE

2.1 Main Objective

To develop an AI health monitoring module tracking vital metrics without wearables while delivering early warning alerts for hypertension and diabetes with 80% predictive accuracy, integrated with fitness and nutrition modules within 14 weeks. This SMART objective enables continuous preventive care for Sri Lanka's NCD crisis affecting 1 in 3 adults.

2.2 Specific Objectives

VitalAI: Intelligent Health Monitoring Module

Objective 1: To design and develop a smartphone-based health tracking system that monitors blood pressure trends, blood sugar patterns, sleep quality, step count, and symptom frequency

without requiring external wearable devices. This objective involves creating intuitive user interfaces for manual data entry of health metrics, supplemented by smartphone sensor integration for step counting using accelerometer data and sleep tracking using phone usage patterns. The system will support both one-time entries and scheduled reminders for regular logging, with data visualization through trends and charts that make patterns easily understandable. Success will be measured by 90% user retention in daily logging during testing and positive usability scores of at least 4 out of 5 in System Usability Scale evaluations with 30 participants across different age groups and technology literacy levels.

Objective 2: To develop AI-powered early warning algorithms that detect rising risk for hypertension and diabetes by analyzing longitudinal health data and lifestyle patterns, delivering alerts 2-4 weeks before traditional symptom onset. This objective involves training machine learning models on retrospective health data to identify patterns preceding NCD diagnosis, incorporating variables including blood pressure variability, blood glucose trends, weight changes, physical activity levels, and sleep quality. The algorithms will generate personalized risk scores and deliver early warnings through in-app notifications and optional SMS alerts in Sinhala and Tamil. Success requires achieving at least 80% predictive accuracy validated against clinical datasets and successful detection of simulated deteriorating patterns in test scenarios.

Objective 3: To implement a multilingual AI-powered symptom checker that provides basic health guidance in Sinhala and Tamil languages through natural language conversation, enabling users to assess common symptoms and receive appropriate recommendations. This objective focuses on developing a conversational interface using Dialogflow or Rasa trained on medical knowledge bases and local healthcare protocols, supporting at least 20 common symptoms including headache, fever, fatigue, chest pain, and digestive issues. The symptom checker will provide triage recommendations, suggest when to seek professional care, and offer self-care guidance while clearly disclaiming that it does not replace medical diagnosis. Success metrics include 85% accuracy in symptom classification and user satisfaction ratings of at least 4 out of 5 for helpfulness and clarity.

Objective 4: To establish bidirectional API integration with FitAI, NutriAI, and MindAI modules enabling comprehensive health data sharing that triggers automated interventions across the ecosystem. This objective involves designing APIs that allow VitalAI to share real-time health updates with other modules while receiving behavioral data that enriches its predictive models. The integration will enable scenarios where deteriorating sleep patterns detected by VitalAI trigger MindAI meditation recommendations, and rising blood glucose automatically adjusts FitAI exercise intensity and NutriAI meal plans. Success requires successful end-to-end data flow in 95% of test transactions with appropriate privacy safeguards including user consent management and data encryption.

Objective 5: To evaluate VitalAI's effectiveness in improving health awareness and early detection through a 6-week user study with 40 participants including individuals with and without existing NCD diagnoses. This objective involves recruiting participants from urban and rural areas, providing them with VitalAI for 6 weeks, and assessing changes in health awareness, logging consistency, and early detection capability. Pre and post-intervention surveys will measure health literacy and perceived control over health conditions. Success requires demonstrating that 80% of participants report increased awareness of their health patterns and that the system successfully identifies at least three cases of concerning trends requiring medical consultation during the study period.

3. METHODOLOGY

3.1 Research Approach and Technical Methods

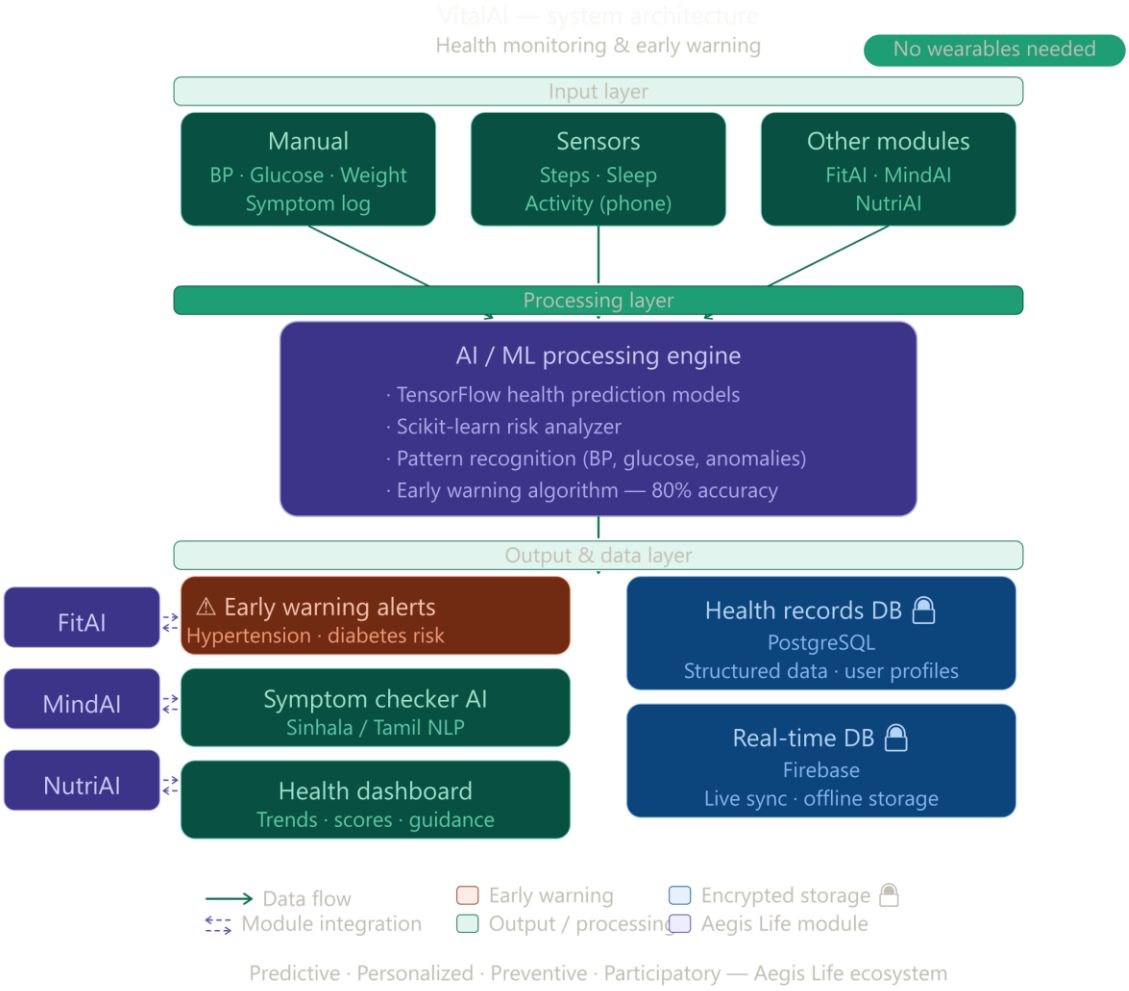
VitalAI will employ a Design Science Research methodology combined with quantitative predictive modeling, appropriate for developing and validating a novel health monitoring artifact while systematically evaluating its predictive accuracy and usability. The research will follow the six-phase Design Science paradigm: problem identification establishing Sri Lanka's NCD monitoring gap, objective definition specifying early warning capabilities, design and development of the VitalAI system, demonstration through functional prototype, evaluation through rigorous testing with target users, and communication of findings. The technical methodology centers on machine learning for predictive health analytics, utilizing Scikit-learn and TensorFlow to develop classification and regression models that identify patterns preceding hypertension and diabetes complications. Algorithms will include Random Forest classifiers for risk stratification, Long Short-Term Memory networks for analyzing temporal sequences in blood pressure and glucose trends, and anomaly detection models identifying deviations from individual baseline patterns. Feature engineering will extract meaningful indicators from user-inputted data including blood pressure variability, glucose trend slopes, weight trajectories, sleep quality metrics, and physical activity levels, combining these into composite risk scores. The symptom checker component will employ natural language processing using Rasa or Dialogflow trained on medical knowledge bases and Sri Lankan healthcare protocols, enabling Sinhala and Tamil language understanding through custom intent recognition and entity extraction. Rule-based triage logic will implement clinical guidelines adapted for local context, providing appropriate recommendations while maintaining clear disclaimers about the system's limitations as a supportive tool rather than medical diagnostic device. All predictive models will be designed for on-device execution where possible, with sensitive health data encrypted and processed locally to maintain privacy, while anonymized aggregate data may be used for model improvement with explicit user consent.

3.2 Tools, Data Collection, and Validation Strategy

Development tools include React Native with Expo for cross-platform mobile development, chosen for consistent user experience across devices and Expo's sensor integration capabilities for step tracking and basic health monitoring. TensorFlow Lite and Scikit-learn enable on-device and cloud-based predictive modeling respectively, with model training conducted using Jupyter notebooks on Google Colab leveraging GPU acceleration. Firebase provides secure backend services for user authentication, encrypted health data storage, and real-time synchronization with FitAI, NutriAI, and MindAI modules through RESTful APIs. PostgreSQL will store structured health records including blood pressure logs and glucose

readings, while MongoDB accommodates flexible symptom check histories. Data collection involves two streams: retrospective clinical datasets obtained through collaboration with local healthcare providers for model training (anonymized and ethics-approved), and prospective user data from 40 participants during validation studies. Inclusion criteria ensure representation of diverse demographics including urban and rural residents, various age groups, and individuals with and without existing NCD diagnoses. Ethical considerations are paramount given sensitive health data: ethics approval will be obtained from SLIIT's research ethics committee, informed consent clearly explains data usage and privacy protections, all data is anonymized at collection, users control data sharing preferences, and the application explicitly disclaims medical diagnostic capabilities while encouraging professional healthcare consultation. Validation strategy includes quantitative evaluation of predictive accuracy using k-fold cross-validation on retrospective datasets, achieving target 80% accuracy for early warning alerts. Usability testing with 40 participants over 6 weeks will assess system usability scale scores, logging consistency, and user satisfaction. Integration testing verifies bidirectional data flow with other modules through 50 test scenarios. Performance metrics include predictive accuracy (precision, recall, F1-score), early warning lead time (days before traditional detection), user retention rates, and System Usability Scale scores benchmarked against industry standards.

4. HIGH-LEVEL SYSTEM ARCHITECTURE



4. Aegis Life employs a three-layer architecture with four integrated AI modules. The Presentation Layer (React Native Expo) connects users through mobile app with camera, offline mode, and notifications. The Application Layer houses FitAI (pose estimation), VitalAI (health monitoring), MindAI (wellness chatbot), and NutriAI (food scanner) connected via Firebase API Gateway. The Bottom Layer provides databases (PostgreSQL, MongoDB) and ML models (TensorFlow, MediaPipe) for intelligent health recommendations

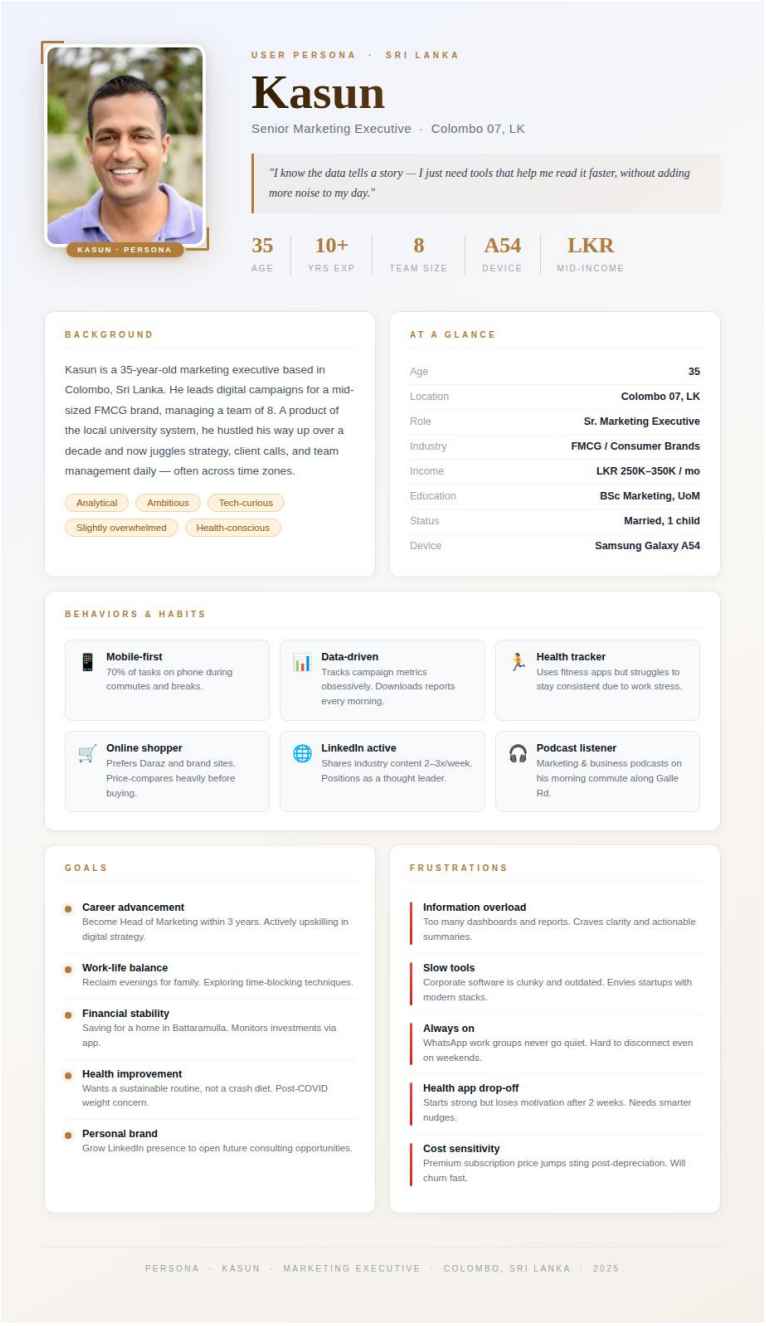
5. USER REQUIREMENTS

- Stakeholder identification
- Functional requirements (numbered)
- Non-functional requirements (security, performance, usability, etc.)

COMMERCIALIZATION PLAN

Target Market and Customer Persona

Kasun is a 35 year old Colombo marketing executive who needs an integrated, affordable wellness app to manage health amidst work stress and information overload



Value Proposition

- Aegis Life offers the first integrated AI-powered wellness platform tailored specifically for Sri Lankans:
- For Fitness Seekers: Real-time exercise form correction using pose estimation, preventing injuries and maximizing workout effectiveness
- For NCD Patients: Early warning system for hypertension/diabetes without expensive wearables, enabling preventive care
- For Mental Wellness: 24/7 AI chatbot in Sinhala/Tamil with culturally-adapted Buddhist meditation practices
- For Nutrition: AI food scanner recognizing Sri Lankan dishes with calorie tracking and personalized meal plans
- Integration Advantage: Unlike standalone apps, all modules share data—a diabetic user's rising blood glucose automatically triggers adjusted workouts and meal plans

Revenue Model

Model	Description
Freemium Subscription	Basic features free (limited workouts, basic tracking). Premium at Rs. 500-800/month for advanced features (real-time form correction, early warnings, full meal planning)
Corporate Wellness Programs	B2B subscriptions for companies (50-500 employees) at Rs. 400/user/month. Addresses 50% workplace burnout rate
In-app Coaching	Commission-based (15-20%) on bookings with fitness coaches and nutritionists
Insurance Partnerships	B2B contracts with insurance companies offering premium discounts for policyholders using the app (reduces their NCD claim costs)
Data Insights (Anonymized)	Aggregated health trends for research institutions and Ministry of Health

Cost Estimation (Annual)

Cost Category	Amount (LKR)
Cloud Services (Firebase, hosting)	18,000
Staff/Developer maintenance (part-time)	12,000
Marketing & User Acquisition	8,000
API & Third-party services	6,000
Customer Support	6,000
Total Annual Operating Cost	50,000

Pricing Strategy

Tier	Price	Target Users	Features
Free	Rs. 0	Budget-conscious users, students	Basic workout videos, manual health logging, limited chatbot, basic meal tracking
Premium Monthly	Rs. 599	Individual professionals, health-conscious users	Full features: real-time form correction, early warnings, unlimited chatbot, personalized meal plans
Premium Annual	Rs. 5,999 (Rs. 500/month)	Committed users	Same as monthly + 2 months free
Corporate	Rs. 400/user/month	Companies (min. 50 users)	All premium + admin dashboard, team challenges, HR reports

Competitive Advantage

Advantage	Aegis Life	Competitors
Integration	All 4 modules connected	Standalone apps (separate fitness, health, mental apps)
Language	Full Sinhala/Tamil support	English only
Wearable-free	Works without expensive devices	Require Fitbit/Apple Watch
Local Context	Sri Lankan foods, Buddhist meditation, local body types	Western content only
Price	Rs. 599/month affordable	Rs. 15,000+/year or Rs. 3,000/session
Health Adaptation	Exercises adapt to your health conditions	Generic workouts only

Intellectual Property Considerations

Our IP strategy encompasses multiple protection layers. We will register "Aegis Life" and logo as trademarks in Sri Lanka while securing copyright protection for source code, UI/UX designs, and original meditation content. Patent exploration will focus on two novel aspects: the integrated multi-module health adaptation algorithm and context-aware exercise form correction for NCD patients. Proprietary ML model training methodologies will remain trade secrets, and clear terms of use will establish that users own their personal data while we utilize anonymized aggregates. We will ensure full compliance with open source licenses for Media Pipe, Tensor Flow, Rasa, and React Native.

Go-to-Market Strategy

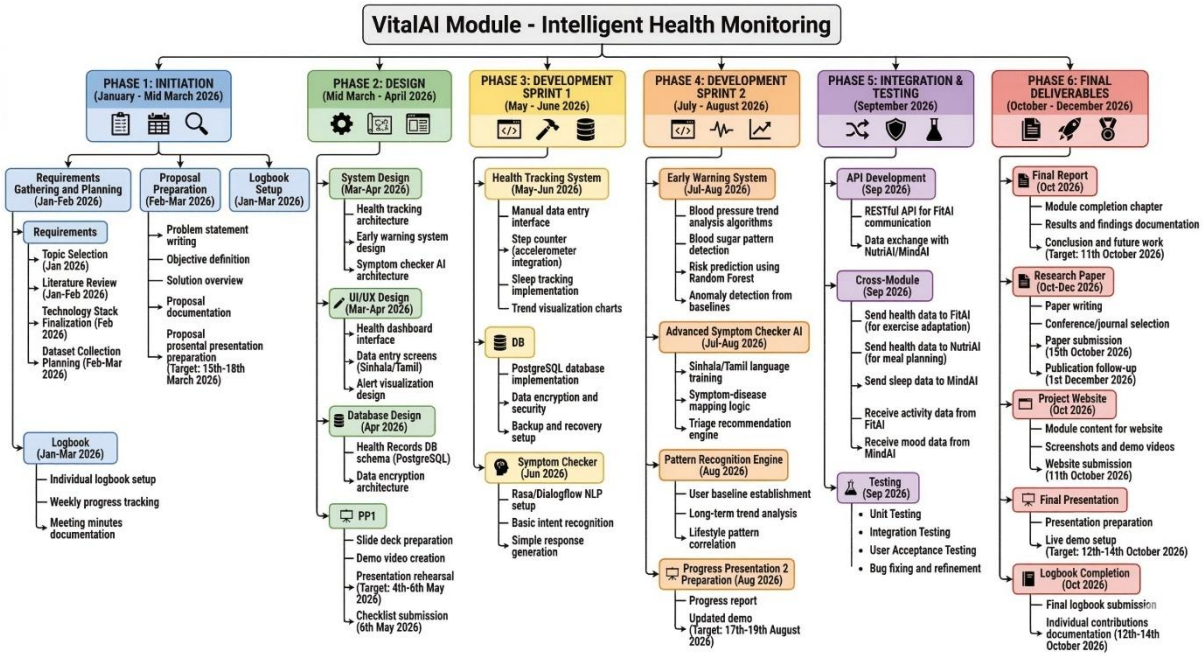
Our first-year rollout follows a phased approach. Months 1-3 involve beta testing with 200 users from universities and corporate partners. Month 4 marks soft launch on Google Play Store, with iOS following in Month 6. Months 5-6 focus on digital marketing campaigns across Facebook, Instagram, and YouTube targeting Sri Lankan audiences. Month 7 initiates corporate pilots with 2-3 companies for wellness programs, followed by insurance partnership negotiations in Month 9. By year-end, we target 10,000 active users with 15% premium conversion.

7. BUDGET AND JUSTIFICATION

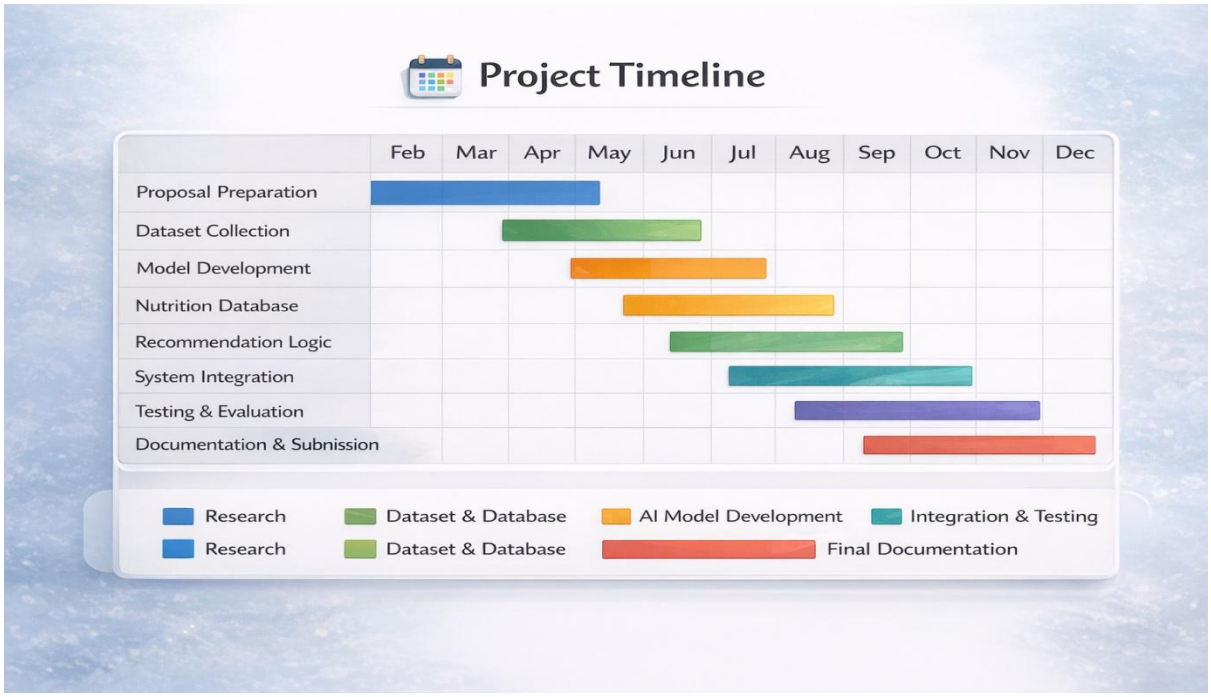
Category	Amount (LKR)
Hardware Costs	31,700
Software Licenses	44,800
Cloud Services	46,500
Human Effort	18,500
Contingency	5,200
TOTAL	146,700

The total project budget is estimated at LKR 146,700, allocated across five main categories to support the 12-month development of the Aegis Life platform.

8. WORK BREAKDOWN STRUCTURE (WBS)



9. GANTT CHART



10.REFERENCES LIST

[1] K. Anand, P. Kumar, and R. Hemanth Kumar, "An AI-Driven Multi-Modal Framework for Exercise Form Detection and Correction in Strength Training," in *2025 6th International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV)*, Tirunelveli, India, 2025, pp. 1-6, doi: 10.1109/ICICV64824.2025.11085771.

[2] "Smartfit: Artificial Intelligence-Powered Exercise Recognition," in *2025 IEEE International Conference on Intelligent Systems and Computing (ICISC)*, Coimbatore, India, 2025, pp. 1-5, doi: 10.1109/ICISC.2025.11033069.

[3] Y. D. De Mel and P. M. Nallapperuma, "Fitness Warrior: Fitness and Nutrition Tracker with Personalized Goal Generation," *SLIIT City UNI Research Symposium ARCSCU 2025*, pp. 77-84, Jul. 2025, doi: 10.54389/BAUC2527.

[4] A. C. Nigam and R. S. Chanda, "Just Let Them Play: An Empirical Demonstration of Customer Retention Through Gamification in Mobile Fitness App," *Lex localis - Journal of Local Self-Government*, Aug. 2025, doi: 10.52152/a6jt5404.

[5] World Economic Forum, "Frontiers in Digital Health: Designing digital health interventions with causal inference and multi-armed bandits," *Strategic Intelligence*, Jun. 2025. [Online]. Available: https://intelligence.weforum.org

[6] M. Shiiba et al., "Comparison of contact diaries and wearable proximity sensors in measuring household contacts in low- and middle-income countries," *medRxiv*, Oct. 2025, doi: 10.1101/2025.10.07.25336904.

[7] C. C. Chu, "AIoT in health care challenges and opportunities," presented at Peking University Computer Science Lecture Series, Beijing, China, Nov. 2023.

[8] "AI Chatbot as a Kurdish Mental Health Clinician: A Digital Mental Health Program," in *eMHIC Congress 2025*, Virtual, Nov. 2025.

[9] "HamRaz: A Culture-Based Persian Conversation Dataset for Person-Centered Therapy Using LLM Agents," *arXiv preprint arXiv:2502.05982*, Feb. 2025.

[10] T. de Azevedo Cardoso et al., "Digital Mental Health Interventions for Adolescents in Low- and Middle-Income Countries: Scoping Review," *Journal of Medical Internet Research*, vol. 26, p. e51376, Oct. 2024, doi: 10.2196/51376.

11. APPENDICES

12. Appendix A: Survey Instruments

Appendix	Title	Description
A-1	User Needs Assessment Survey	Questionnaire to understand fitness app requirements from 100+ Sri Lankan users
A-2	Exercise Preference Survey	Survey identifying 15 most common exercises in Sri Lankan home settings
A-3	Post Testing Feedback Form	User satisfaction survey for beta testers after using
A-4	System Usability Scale (SUS) Questionnaire	Standardized usability measurement tool (Sinhala/Tamil translated)

13. Appendix B: Detailed Diagrams

Appendix	Title	Description
B-1	Component Specific Architecture	Detailed diagram showing all sub components and data flow
B-2	Pose Estimation Pipeline Diagram	MediaPipe landmark detection process with 33 keypoints

14. Appendix C: Preliminary UI Mockups

Appendix	Title	Description
C-1	Home Screen Mockup	Main dashboard with exercise recommendations (Sinhala/Tamil version)
C-2	Exercise Selection Screen	List of 15 exercises with video thumbnails
C-3	Pose Estimation Camera View	Camera screen with skeleton overlay during exercise

15. Appendix D: Risk Analysis

Appendix	Title	Description
D-1	Technical Risk Assessment Matrix	Risks: pose estimation accuracy <85%, latency >300ms, model overfitting
D-2	Mitigation Strategies Document	Backup plans for each identified technical risk
D-3	Project Timeline Risk Analysis	Risks affecting proposal (Mar 15), PP1 (May 4-6), PP2 (Aug 17-19), Final (Oct 11-14)
D-4	Data Privacy Risk Assessment	Risks related to storing exercise videos and user data